

Ganesh Gupta

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EXPERIENCE

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Built infrastructure in **Go**, **C++**, and **Python** for **GPU** scheduling and fault-tolerant placement across multi-cloud fleets on **AWS**, **GCP**, and **CoreWeave**.
- Kept long-running distributed jobs alive through node loss, **stragglers**, and contention on **InfiniBand/RoCE** clusters with **Slurm** scheduling and fast restarts.
- Built **CPU-driven** agentic workloads with tool use and code execution across **Docker** and **AWS Lambda**, running alongside GPU inference.
- Built **Prometheus/Grafana** observability and **autoscaling** for **vLLM** inference serving, correlating compute and network telemetry to catch regressions early.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Optimized LLM serving with **CUDA/Triton** kernel work, **speculative decoding**, and **TensorRT-LLM** alongside **vLLM** on **Kubernetes**.
- Owned **ML infrastructure** end to end, from model deployment to evaluation and debugging, cutting **p99 latency 420ms to 140ms** across **50K+ daily requests**.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- **Profiled compute and memory bottlenecks** across **Python/C++** services and optimized batching and **concurrency**, raising throughput **2.4x** and cutting memory **27%**.
- Built and operated **25+ production microservices** for **1M+ users** on **AWS** (EC2, RDS, SQS), sustaining **99.9% uptime** over four years.

PROJECTS

Distributed KV Store

github.com/ganeshhgupta/distributed-kv-store

- **Raft-consensus** distributed key-value store with **linearizable reads**, leader election, and **log compaction**.

RootCause AI

github.com/ganeshhgupta/rootcause-ai

- **Multi-agent** incident-diagnosis system with **tool-use** chains that collects evidence from logs and metrics and synthesizes root-cause reports.

Watch-your-LLM

github.com/ganeshhgupta/watch-your-llm

- **LLM evaluation** platform with rubric-based scoring, drift detection, and latency tracking across model versions; **FastAPI**, **PostgreSQL**, regression alerting.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Jul 2016 – Jul 2020

SKILLS

Languages: Python, C++, Go, CUDA, SQL

Distributed Training & RL: Training-Inference-Rollout Loops, Reinforcement Learning, Fault Tolerance, Elastic Scaling, PyTorch

GPU & HPC: CUDA/Triton, vLLM, TensorRT-LLM, InfiniBand/RoCE, Slurm, Multi-GPU

Cloud & DevOps: AWS, GCP, Kubernetes, Docker, Terraform, CI/CD, Prometheus, Grafana

Fundamentals: Data Structures & Algorithms, Operating Systems, Database Systems, Computer Networks, HTML, CSS

Certifications: AWS Certified Developer Associate, Google Cloud Professional Machine Learning Engineer